Observation:

Focal length of the convex lens (By distance object method) is (f) = ……cm

2f = …….cm

S.No	Size of the Image	Position of the object	Distance between the object and the lens (u) cm	Distance between the screen and the Lens (v) cm	Focal length of convex lens $f = \frac{uv}{(u+v)}$ **cm**
1	Diminished	u > 2f			
2					
3	Same size	u = 2f			
4	Magnified	u < 2f			
5					

Result:

The focal length of the given convex lens

1. By distance object method f = ……cm
2. By 'uv' method f = … ……..cm

3. DETERMINATION OF RESISTIVITY

Objective:

To determine the resistivity of the material of the given coil of wire.

Equipment required:

A coil of wire, screw gauge, a metre scale, battery, key, ammeter, voltmeter, rheostat and connecting Wires.

Formula:

The resistivity of the material of the coil of wire is

$$\rho = \left(\frac{A}{L}\right)R \quad (in\ ohm\ metre)$$

Where A is the area of cross section of the wire (m^2)

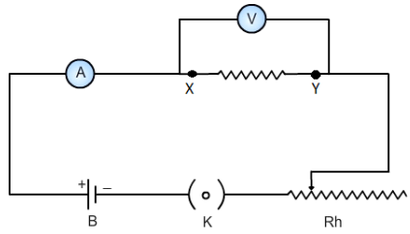
L is the length of the coil of wire (m)

` R is the resistance of the coil of wire (ohm)

Circuit Diagram:

Procedure:

- Connect the battery, ammeter, given wire, rheostat and key in series, as shown in the circuit diagram.
- Connect the voltmeter in parallel to the unknown resistor.



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